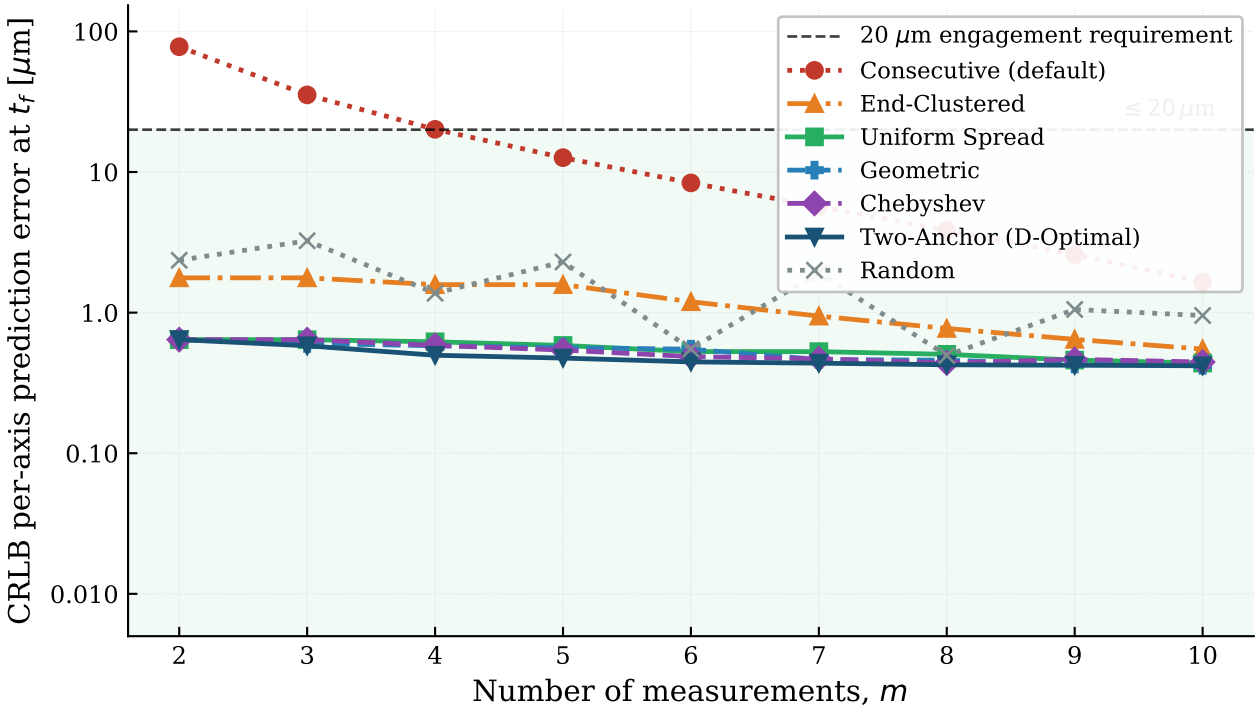
Effect of Sensor Placement on Tracking Precision
Cramér–Rao Lower Bound ($\sigma_{meas} = 5\,\mu$m, $\tau_{latency} = 3$ ms)
CRLB per-axis prediction error at t_f [μm]
Number of measurements, m
100
10
1.0
0.10
0.010
2
3
4
5
6
7
8
9
10
20 μm engagement requirement
Consecutive (default)
End-Clustered
Uniform Spread
Geometric
Chebyshev
Two-Anchor (D-Optimal)
Random
$\leq 20\,\mu$m